

MID QUARTER REPORT

Team 20

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[Medical Insurance Cost Prediction](#)

DATASET DESCRIPTION

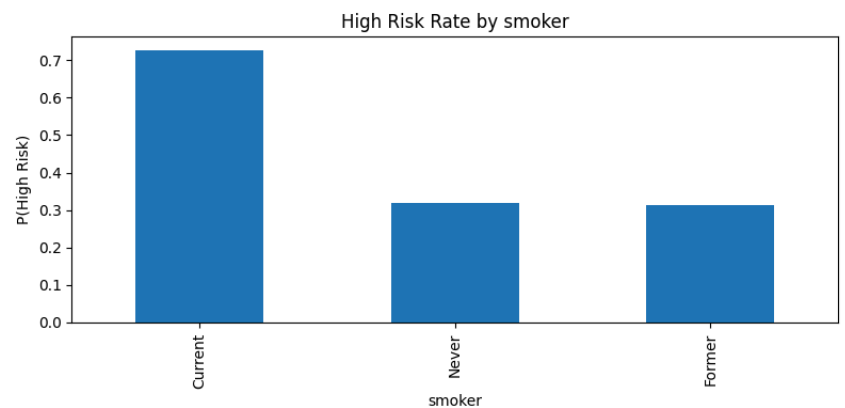
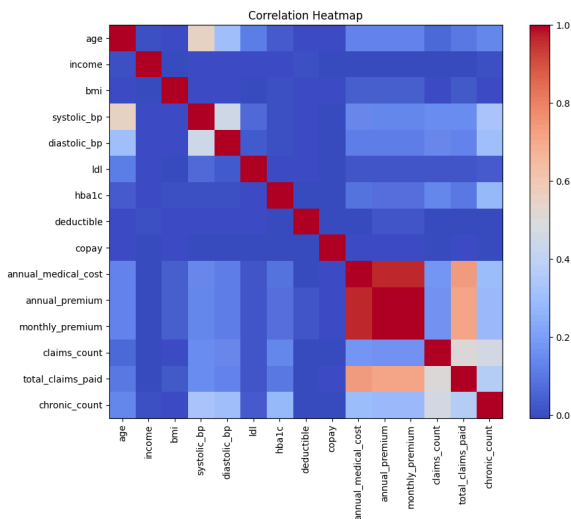
1. Our dataset comes from Kaggle. It consists of 100,000 rows and 54 columns and covers a wide range of patient info. Within the dataset there's basic demographic data such as age, sex, region, education level, and health data including BMI, blood pressure, HbA1c, and LDL cholesterol. It also tracks chronic conditions including hypertension, diabetes, asthma, COPD, heart disease, cancer history, kidney disease, liver disease, arthritis, and mental health status. On the insurance side, we have plan type, network tier, deductible, copay, and policy term length. There's also medical usage data such as how many claims someone filed, total claims paid out, and procedure counts for things like imaging, surgery, physical therapy, consultations, and lab work. Finally there's financial info like income, annual medical cost, and premiums.
2. The dataset has a built-in label called `is_high_risk` that marks about 36.8% of patients as high-risk and 63.2% as low-risk. Only one column, `alcohol_freq`, has missing values and that's at about 30%. There are zero duplicate rows.

EXPLORATORY DATA ANALYSIS

1. We ran a full EDA to get a feel for the data before building any models. We looked at how each feature is distributed on its own, how features relate to the risk label, and how features correlate with each other.
2. Most of the continuous variables like age, BMI, and blood pressure look roughly normal. The financial columns like income, medical costs, and claims are skewed right with some big outliers, which makes sense for insurance data since a small number of people tend to have really high costs.
3. The most useful part of the EDA was comparing features across the two risk groups. Age was the biggest differentiator by far. Low-risk patients average about 41 years old while high-risk patients average around 59. Chronic condition count was also a strong signal, with low-risk patients having 0.38 conditions on average versus 1.32 for high-risk. Systolic blood pressure showed a clear gap too, 113 for low-risk vs 127 for high-risk. Claims count and annual medical cost both showed similar patterns with high-risk patients having roughly double the values.
4. For categorical features, smoking status stood out the most. Current smokers had a 73% chance of being high-risk, while former and never smokers were both around 31-32%. That's a huge gap. On the other hand, stuff like sex, region, whether someone lives in an urban or rural

area, plan type, network tier, and employment status barely made any difference in risk rates. Those features had almost identical high-risk percentages across all their categories.

5. The correlation heatmap showed that annual medical cost, annual premium, and monthly premium are all strongly correlated with each other, which makes sense since premiums are based on expected costs. Age also correlates with blood pressure and chronic conditions, which you'd expect since health tends to decline with age.



[FIGURE 1: Correlation Heatmap]

Shows relationships between all the numeric features. Cost and premium variables cluster together, and age-related health metrics correlate with each other.

[FIGURE 2: High Risk Rate by Smoker]

Shows that current smokers are way more likely to be high-risk (73%) compared to former and never smokers (around 31-32%).

FEATURE SELECTION

1. We split the features into four groups for preprocessing. The 23 numeric features like age, income, BMI, and blood pressure get median imputation for any missing values and then standard scaling so they're all on the same scale. The 10 categorical features like sex, region, education, and smoker status get filled with the most common value if missing and then one-hot encoded. The 12 binary features for chronic conditions and the 5 procedure count features just get passed through as-is since they're already in a usable format. We set all of this up using scikit-learn's ColumnTransformer and Pipeline so there's no data leakage between the training and test sets.

2. Even though some features like deductible, copay, and region didn't show much predictive power on their own in the EDA, we kept them in the model anyway. They might still be useful in combination with other features, and the regularization in our models will naturally downweight anything that isn't helping.

LITERATURE REVIEW

1. We looked at existing work on medical insurance cost prediction to see what's been done before. Thamarai and Malarvizhi (2020) used linear regression, random forest, and gradient boosting to predict insurance charges and found that smoking status and BMI were the biggest predictors. But their dataset was only about 1,300 records with 7 features, which is way smaller than what we're working with.

2. Boodhun and Jayabalan (2018) worked on risk prediction in health insurance using logistic regression and decision trees with clinical features similar to ours. They stressed the importance of feature engineering and regularization when dealing with lots of medical features, which lines up with our approach.

3. Our project is different from these past studies in a few ways. First, our dataset is much bigger at 100,000 patients versus the typical few thousand. Second, we have 54 features covering clinical, financial, and utilization data instead of just a handful. Third, we're doing both regression to predict costs and classification to predict risk and insurance quality, rather than just one task.

EARLY PLAN FOR MODEL DEVELOPMENT

1. We've started building models on two tracks. For regression, we trained Linear Regression and Ridge Regression to predict annual medical cost. We used an 80/20 train-test split, pulled out another 20% for validation, and then ran 3-fold cross-validation to tune hyperparameters. Ridge came out on top with an R-squared of 0.966 and RMSE of about 575, which means it's explaining about 97% of the variance in medical costs. The best alpha value was 0.01.

2. For classification, we set up logistic regression with both L1 and L2 regularization using the SAGA solver. The goal here is to classify patients into low, medium, and high insurance quality tiers. We derived this target from a mix of plan type, network tier, deductible, and premium data. Our evaluation methods include F1-macro, Matthews Correlation Coefficient, and ROC-AUC. Next up we need to actually run the classification pipeline end to end, add Random Forest and Gradient Boosting classifiers, and then compare everything.